

Yet another promising technology on the horizon is HAMR (Heat Assisted Magnetic Recording). A technology primarily pushed by Seagate, HAMR uses thermal lasers to heat the metal alloy-based platter to record data. The benefit is by doing so single bits of data can be stored in much smaller areas than with current HDD technology resulting in a much higher areal density. We could very well see 30 TB hard drives based on HAMR technology post 2010.

Flash drives are as portable as storage can get. They're also ultra-convenient and dirt-cheap thanks to several recent price reductions. A year or so ago, a 512 MB USB drive cost Rs 2,000. Now a 4 GB drive (4096 MB) is available for just under the same amount! Just in time for Vista's ReadyBoost. However, Flash prices aren't viable for replacing hard drives any time soon. We see Flash and HDDs go hand in hand for a few years, with Flash taking over most immediate storage and retrieval situations and hard drives kicking in for situations where more space is required.

HD-DVD is also on the horizon; featuring up

to three times the conventional storage per layer (15 GB as opposed to 4.7GB). At CES 2007, Toshiba unveiled 17-GB-per-layer HD-DVDs, and demonstrated a triple-layer (51 GB) disc. HD standards are much more rigorous for both audio and video—stereo sound at 24bit/192 KHz or eight channel sound at 24bit/96KHz. The other high-density optical format, Blu-ray, is the main competition to HD-DVD, as you probably know. Remember, the format war is still on.

Further on the horizon, we have HVD—Holographic Versatile Disc, a technology still in its infancy. Much research is being done but when HVD emerges it should make HD-DVD and BluRay seem like the floppies of old—with around 3.9 TB of data on a single disc!

This year should bring out the best in terms of pricing of current-generation storage technologies. We've also seen some interesting emergences like the use of 1.8-inch solid state drives for laptop storage (currently in 16 and 32 GB capacities). The next few years should see larger flash drives compete with existing HDD technology. May the hoarding continue! ☒

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Printing Better

Printers have been workhorses we all love to flog, as stacks of A4 paper lying around any good office will testify to. Advancements in inks and printers don't just relate to faster prints: printers are now being developed to print on a variety of media. Epson, for example, has developed printers specifically designed to print on certain surfaces, and they've demonstrated this by printing on diverse materials like canvas, a variety of fabrics, and even Papier Maché.

Imagine famous works of art being recreated by printing on oil canvas! Equally intriguing is Epson printer's ability to print on a variety of fabrics and other materials which are then used to create unique decors—very suitable for banquet halls, board rooms, and such. The main advantage is, the look and the design theme can be changed by simple swapping rather than having a costly redesign.

Epson has also developed printers that allow capturing of live TV screens and printing them out. Imagine printing out

recipes, coupons, and other snippets as they appear on your TV channels! Perhaps the biggest innovation is the development of a metal compound containing ink that can actually be used to print directly on PCBs—no more wastage in the fabrication process!

Nearly all these B2B technology solutions are in the market's fish-pond; what's left to be seen is how many bite...

3D printing has been possible for awhile now, and industries have had access to these \$1, 00,000 printers for over a decade. IdeaLab one of the companies in the 3D printing game has plans for a sub-\$5,000 (Rs 2.25 lakh!) 3D printer for the home space very soon. 3D printers create 3D objects by melting plastic particles. IdeaLab has come up with a much cheaper and compact solution that uses a much lower intensity halogen bulb to melt nylon powder. Their vision—a 3D printer in each home...Imagine being able to print your very own toothbrush, comb, and razor!

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